Introduction to Data Science and Engineering
- Bayes theorem

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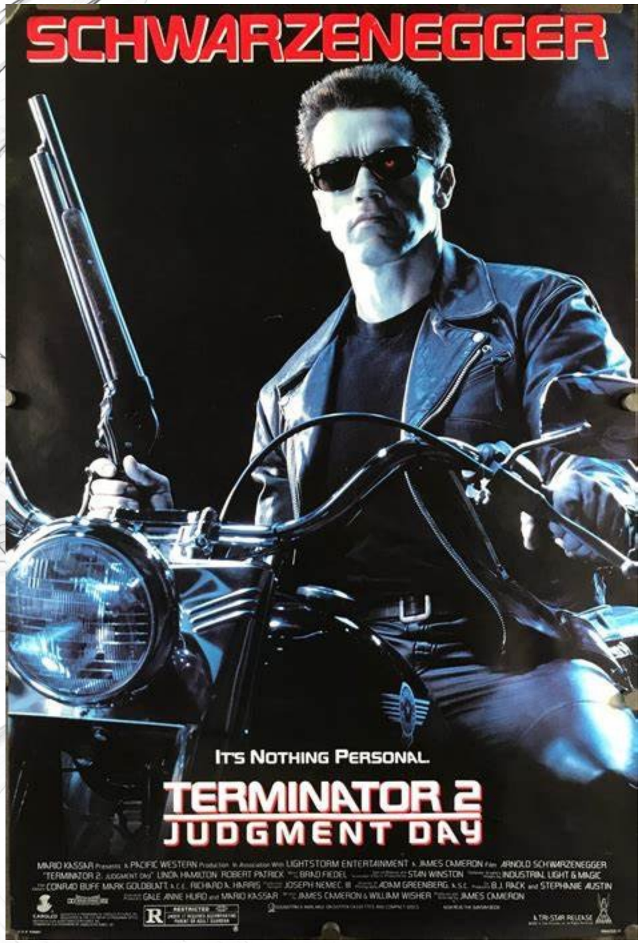


https://arxiv.org/pdf/2303.16200.pdf

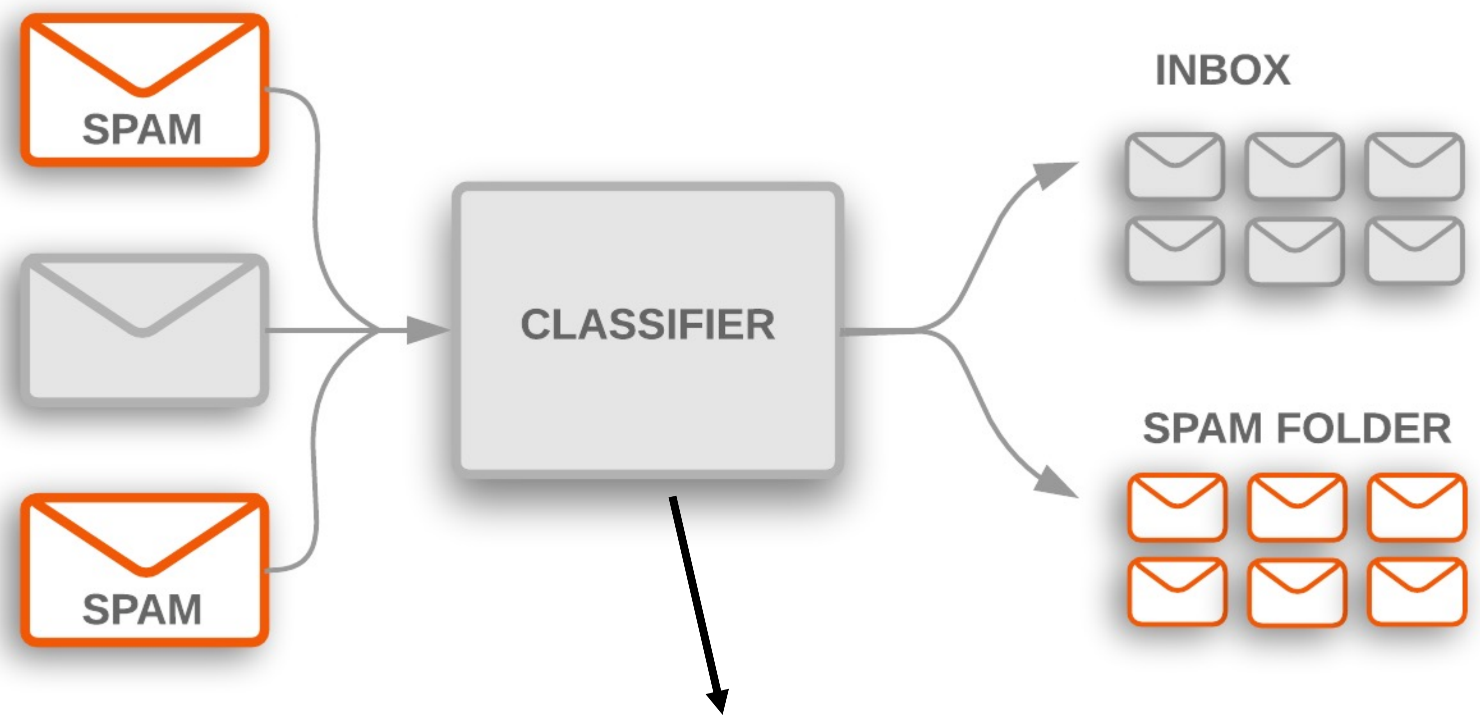
Natural Selection Favors AIs over Humans

Dan Hendrycks
Center for AI Safety

Abstract

For billions of years, evolution has been the driving force behind the development of life, including humans. Evolution endowed humans with high intelligence, which allowed us to become one of the most successful species on the planet. Today, humans aim to create artificial intelligence systems that surpass even our own intelligence. As artificial intelligences (AIs) evolve and eventually surpass us in all domains, how might evolution shape our relations with AIs? By analyzing the environment that is shaping the evolution of AIs, we argue that the most successful AI agents will likely have undesirable traits. Competitive pressures among corporations and militaries will give rise to AI agents that automate human roles, deceive others, and gain power. If such agents have intelligence that exceeds that of humans, this could lead to humanity losing control of its future. More abstractly, we argue that natural selection operates on systems that compete and vary, and that selfish species typically have an advantage over species that are altruistic to other species. This Darwinian logic could also apply to artificial agents, as agents may eventually be better able to persist into the future if they behave selfishly and pursue their own interests with little regard for humans, which could pose catastrophic risks. To counteract these risks and Darwinian forces, we consider interventions such as carefully designing AI agents' intrinsic motivations, introducing constraints on their actions, and institutions that encourage cooperation. These steps, or others that resolve the problems we pose, will be necessary in order to ensure the development of artificial intelligence is a positive one.[1]

Email spam detection



$$\Pr(\text{spam}|\text{words}) = \frac{\Pr(\text{words}|\text{spam})\,\Pr(\text{spam})}{\Pr(\text{words})}$$

Introducing Bayes' theorem

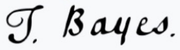
$$P(A|B) = \frac{P(B|A) * P(A)}{P(B)}$$

Thomas Bayes

Portrait used of Bayes in a 1936 book,[1] but it is doubtful whether the portrait is actually of him.[2] No earlier portrait or claimed portrait survives.

Born	c. 1701 London, England
Died	7 April 1761 (aged 59) Tunbridge Wells, Kent, England
Residence	Tunbridge Wells, Kent, England
Nationality	English
Known for	Bayes' theorem

Signature



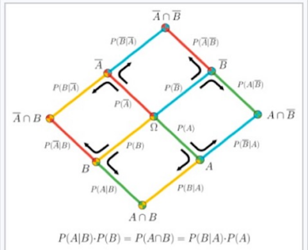
Statement of theorem [edit]

Bayes' theorem is stated mathematically as the following equation:[2]

$$P(A \mid B) = \frac{P(B \mid A)\, P(A)}{P(B)},$$

where A and B are events and $P(B) \neq 0$.

- $P(A)$ and $P(B)$ are the probabilities of observing A and B without regard to each other.
- $P(A \mid B)$, a conditional probability, is the probability of observing event A given that B is true.
- $P(B \mid A)$ is the probability of observing event B given that A is true.



Visualization of Bayes' theorem by the superposition of two decision trees

History [edit]

Bayes' theorem was named after the Reverend Thomas Bayes (1701–1761), who studied how to compute a distribution for the probability parameter of a binomial distribution (in modern terminology). Bayes' unpublished manuscript was significantly edited by Richard Price before it was posthumously read at the Royal Society. Price edited[3] Bayes' major work "An Essay towards solving a Problem in the Doctrine of Chances" (1763), which appeared in "Philosophical Transactions,"[4] and contains Bayes' Theorem. Price wrote an introduction to the paper which provides some of the philosophical basis of Bayesian statistics. In 1765 he was elected a Fellow of the Royal Society in recognition of his work on the legacy of Bayes.[5][6]

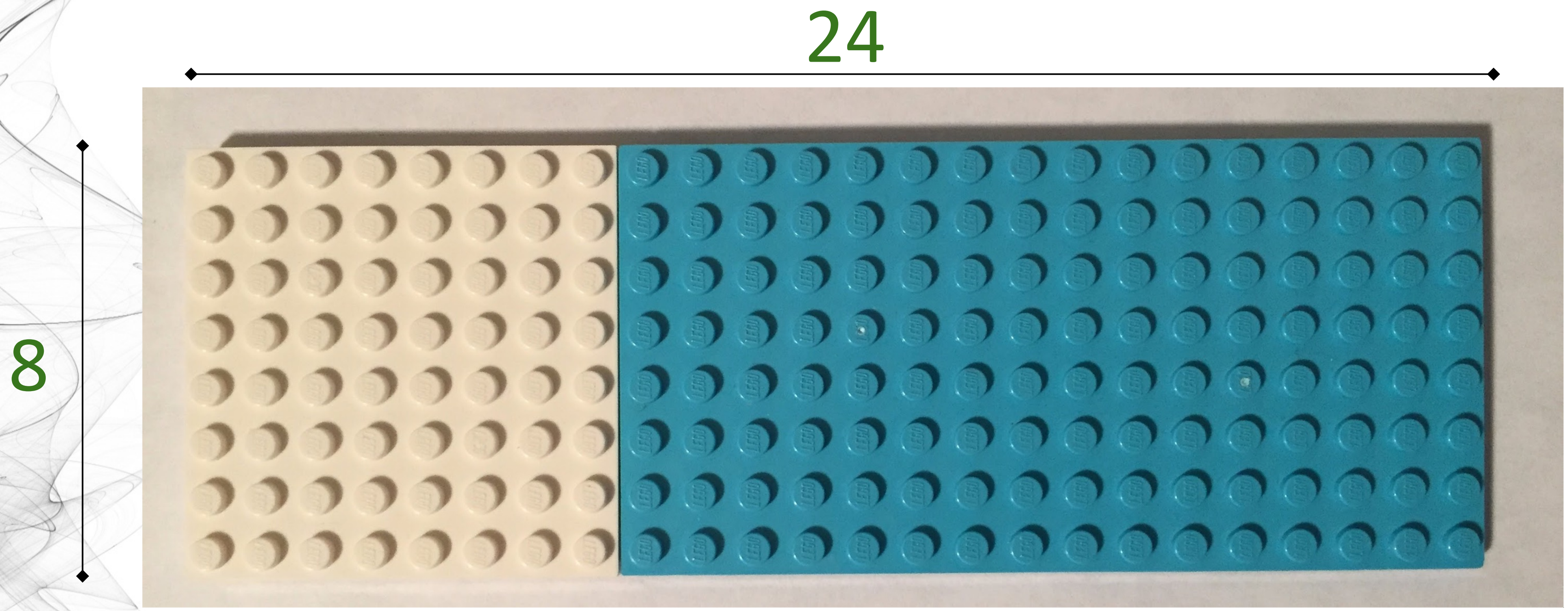
The French mathematician Pierre-Simon Laplace reproduced and extended Bayes' results in 1774, apparently quite unaware of Bayes' work.[7][8] The Bayesian interpretation of probability was developed mainly by Laplace.[9]

Stephen Stigler suggested in 1983 that Bayes' theorem was discovered by Nicholas Saunderson, a blind English mathematician, some time before Bayes;[10][11] that interpretation, however, has been disputed.[12] Martyn Hooper[13] and Sharon McGrayne[14] have argued that Richard Price's contribution was substantial:

> By modern standards, we should refer to the Bayes–Price rule. Price discovered Bayes' work, recognized its importance, corrected it, contributed to the article, and found a use for it. The modern convention of employing Bayes' name alone is unfair but so entrenched that anything else makes little sense.[14]

https://en.wikipedia.org/wiki/Thomas_Bayes

Bayes' theorem with Legos



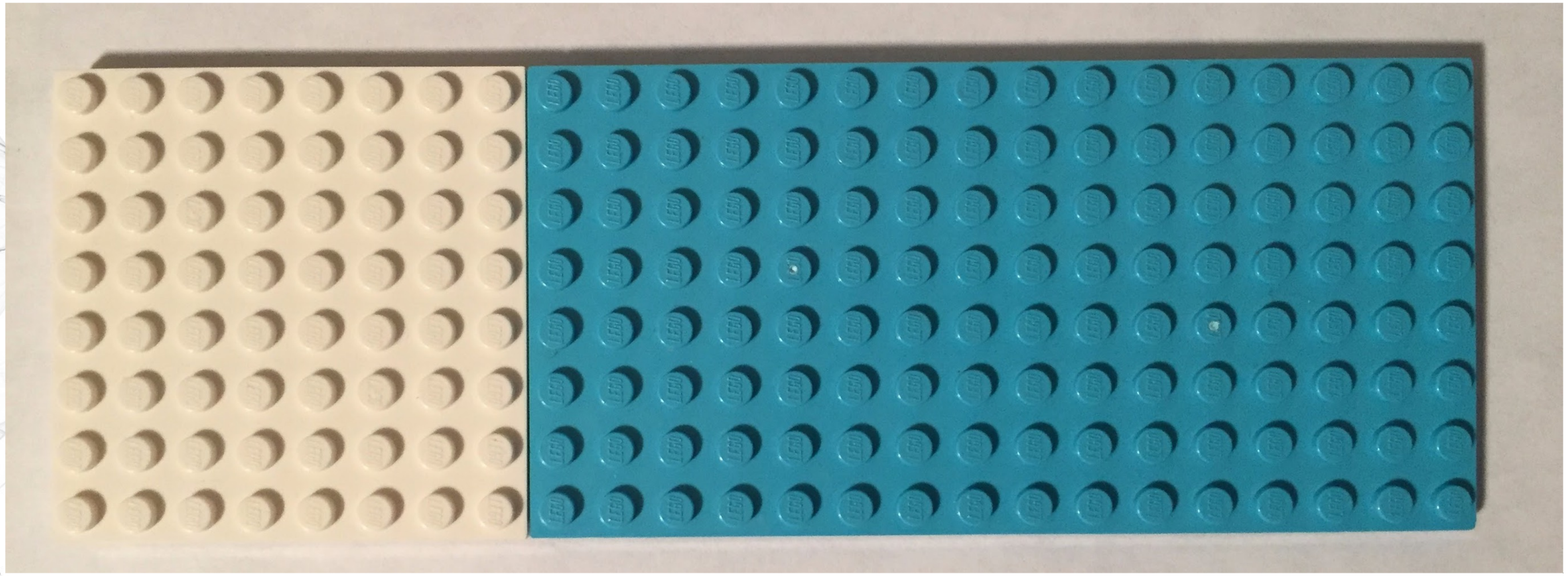
8x24 = 192 pegs, 64 are white, 128 are blue.

P(white) = 64 / 192 = 0.33

P(blue) = 128 / 192 = 0.67

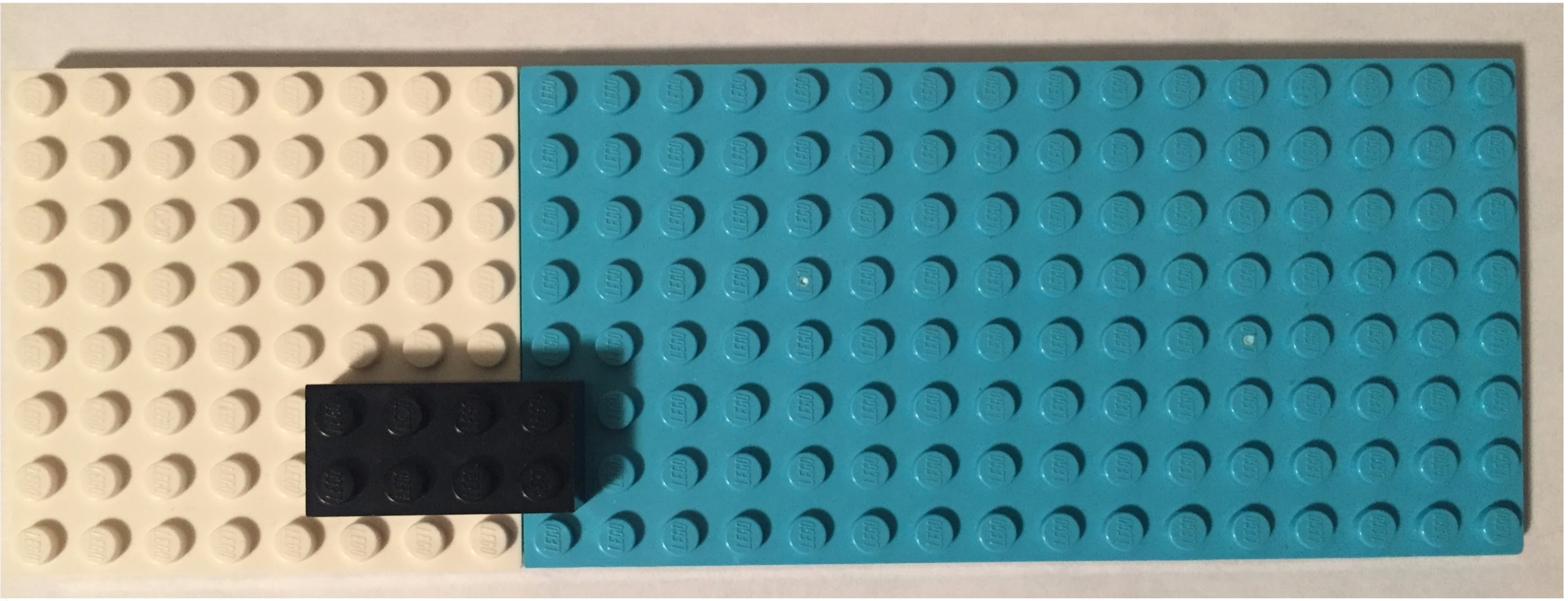
Inspired by https://www.countbayesie.com/blog/2015/2/18/bayes-theorem-with-lego

Our entire probability "space" add up to 1.



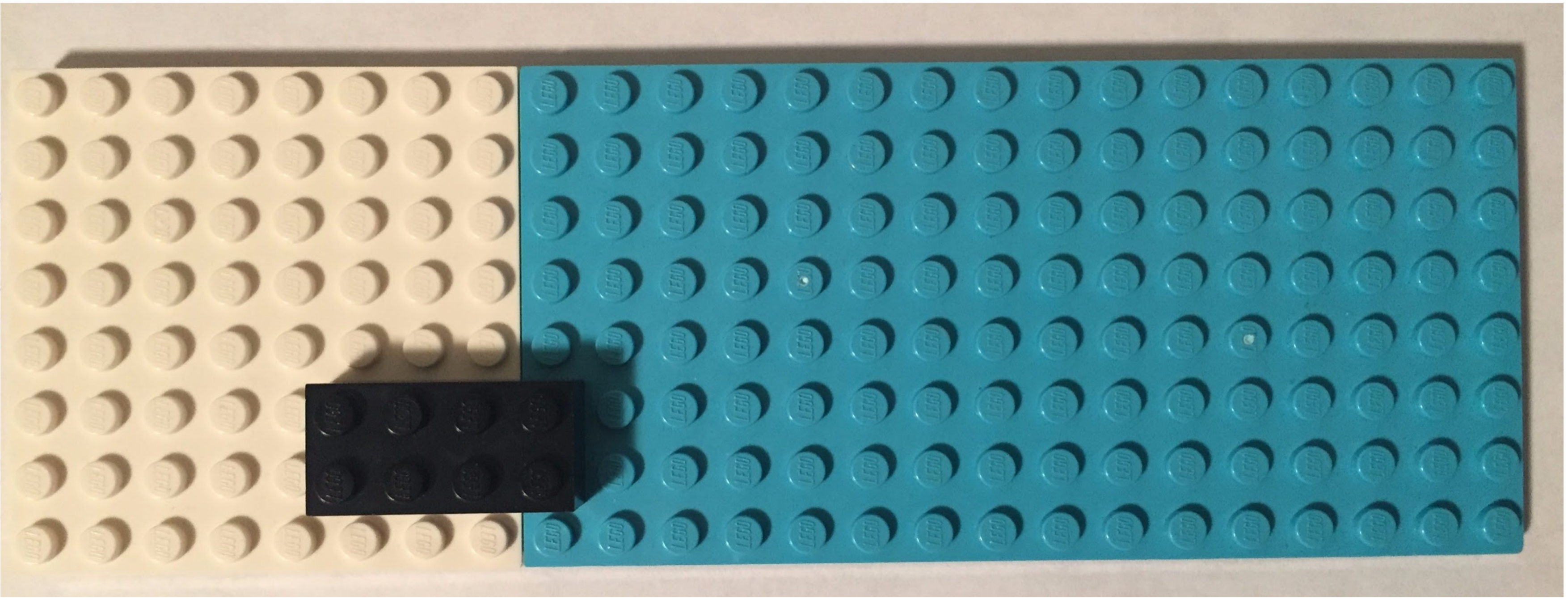
P(white) + P(blue) = 1

What is the probability of black?



P(black) = 8 / 192 = 0.04166

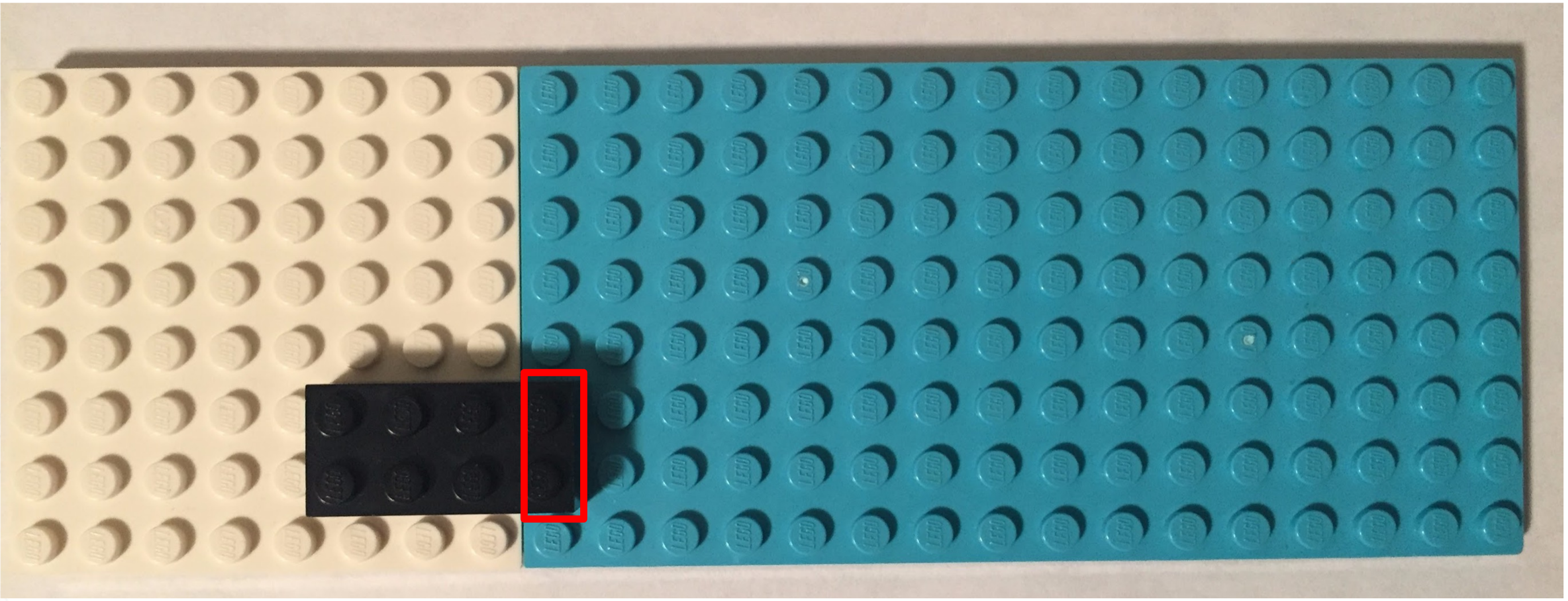
But wait, probability space should always = 1



P(white) + P(blue) + P(black) = 1.042, weird!
What are we missing?

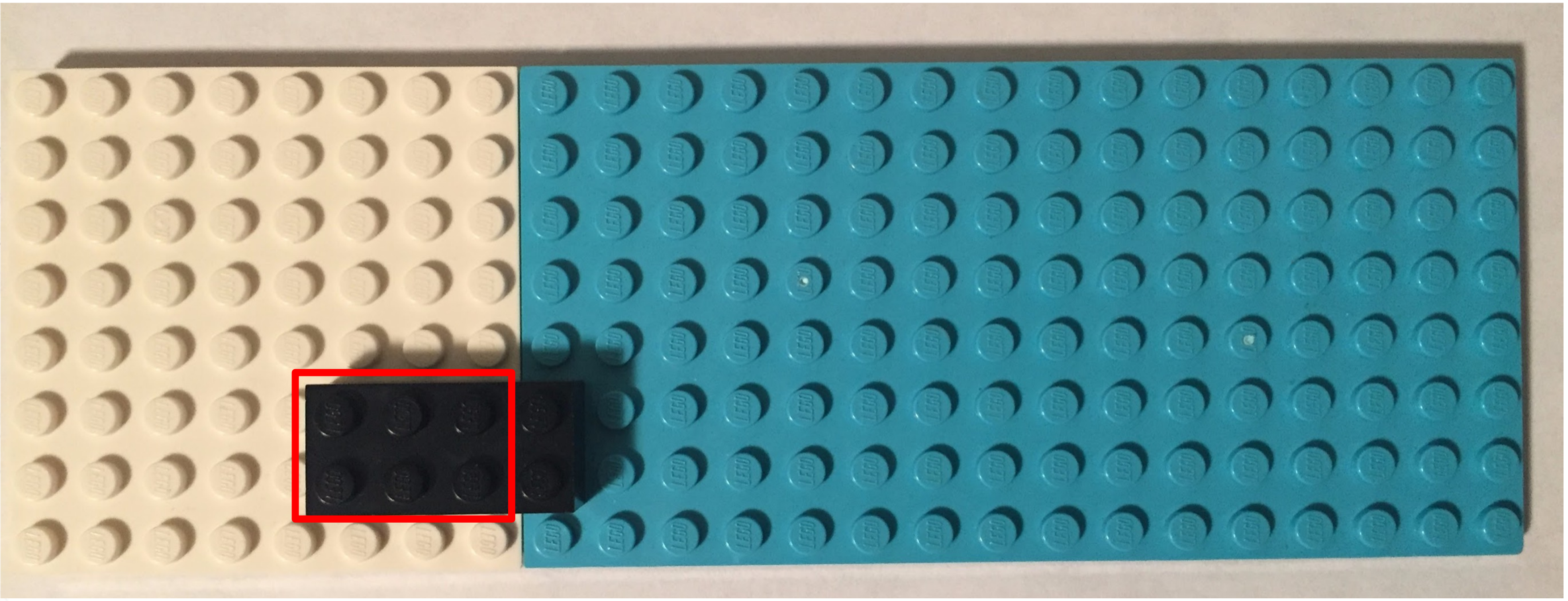
P(black) is <u>conditional on</u> P(white) and P(blue)
So the total space is still P(white) + P(blue)

P(black | blue): "probability of black given that we are on a blue peg"



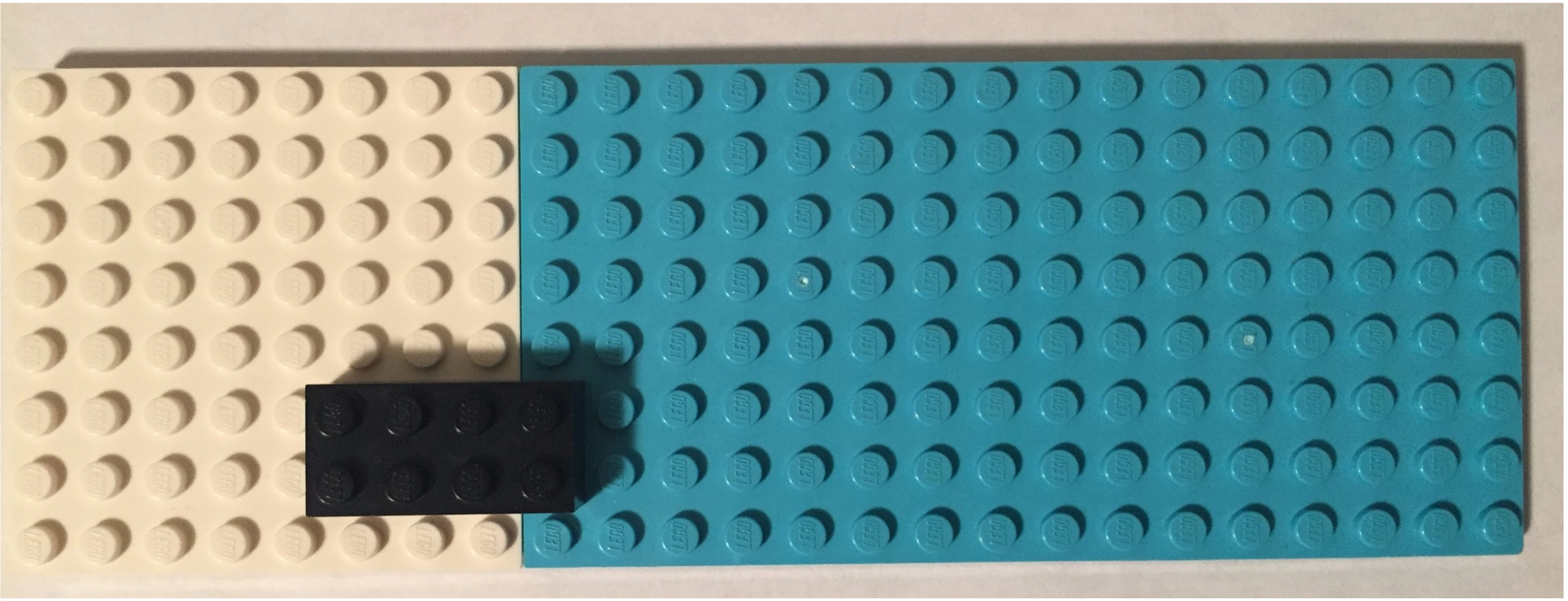
P(black | blue) = 2 / 128 = 0.015625

P(black | white): "probability of black given that we are on a white peg"



P(black | white) = 6 / 64 = 0.09375

But what about the P(blue | black)?



P(blue | black) = 2 / 8 = 0.25

These intuitions can be formalized with Bayes' theorem.

Making sense of Bayes' theorem

aka posterior probability

aka observations

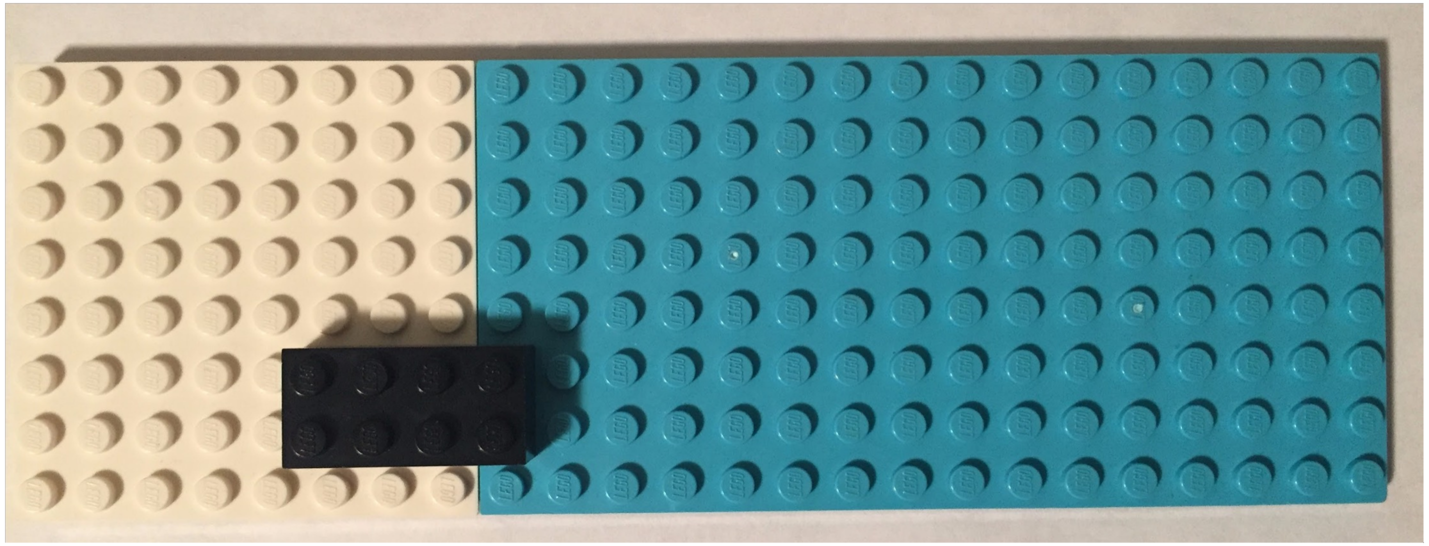
aka prior probability

$$P(A|B) = \frac{P(B|A) * P(A)}{P(B)}$$

aka prior probability

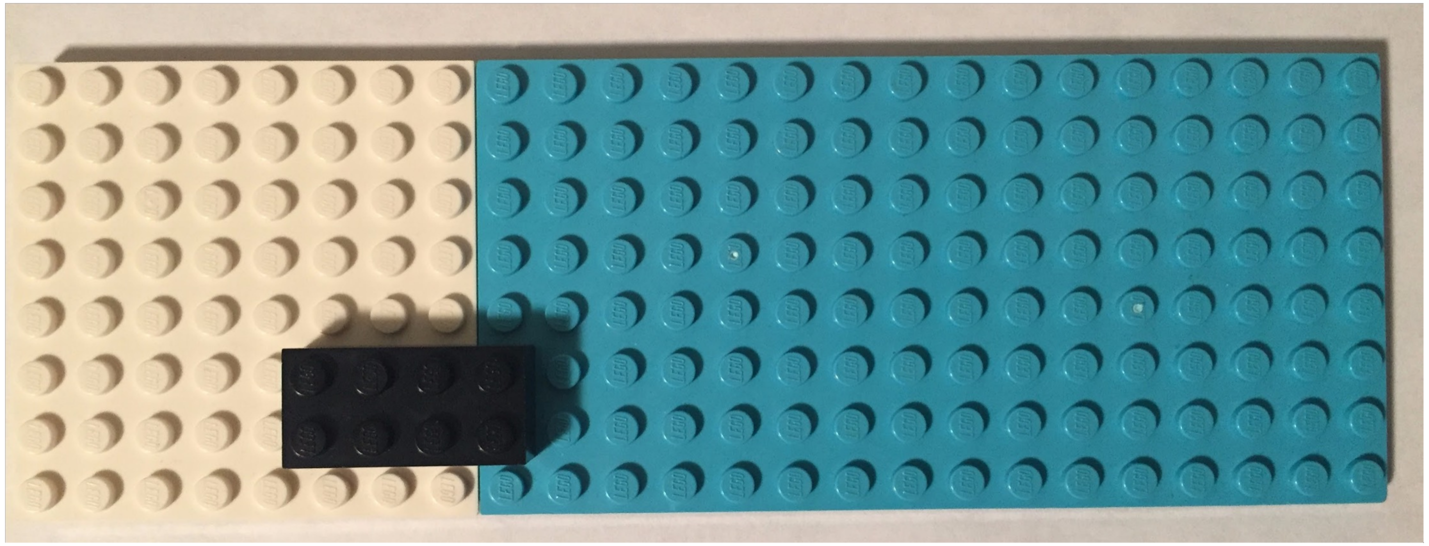
Bayes' theorem

$$P(\text{black}|\text{white}) = \frac{P(\text{white}|\text{black}) * P(\text{black})}{P(\text{white})}$$



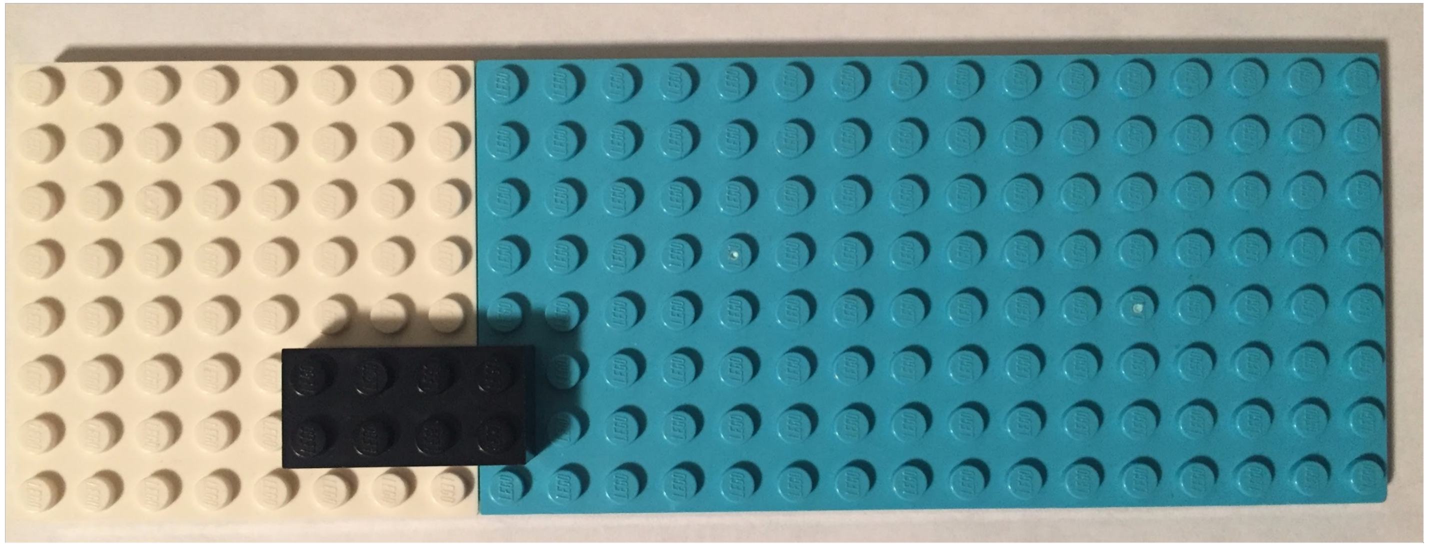
Bayes' theorem

$$P(\text{black}|\text{white}) = \frac{0.75 * 0.04166}{0.33}$$



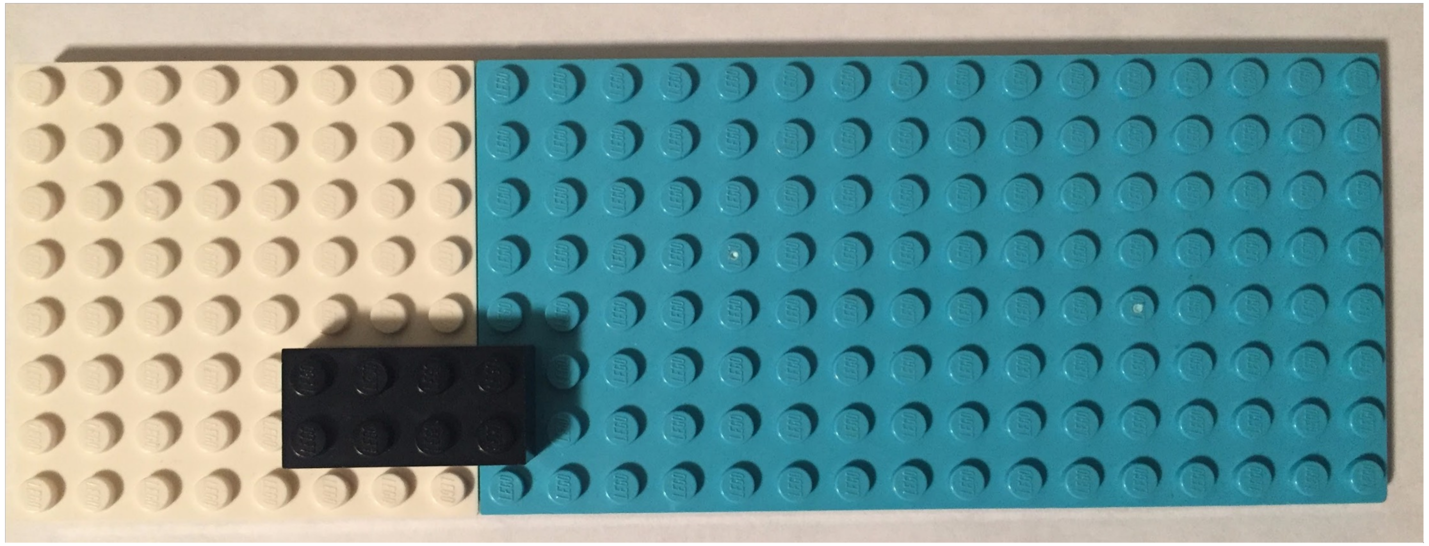
Bayes' theorem

P(black|white) = 0.09375



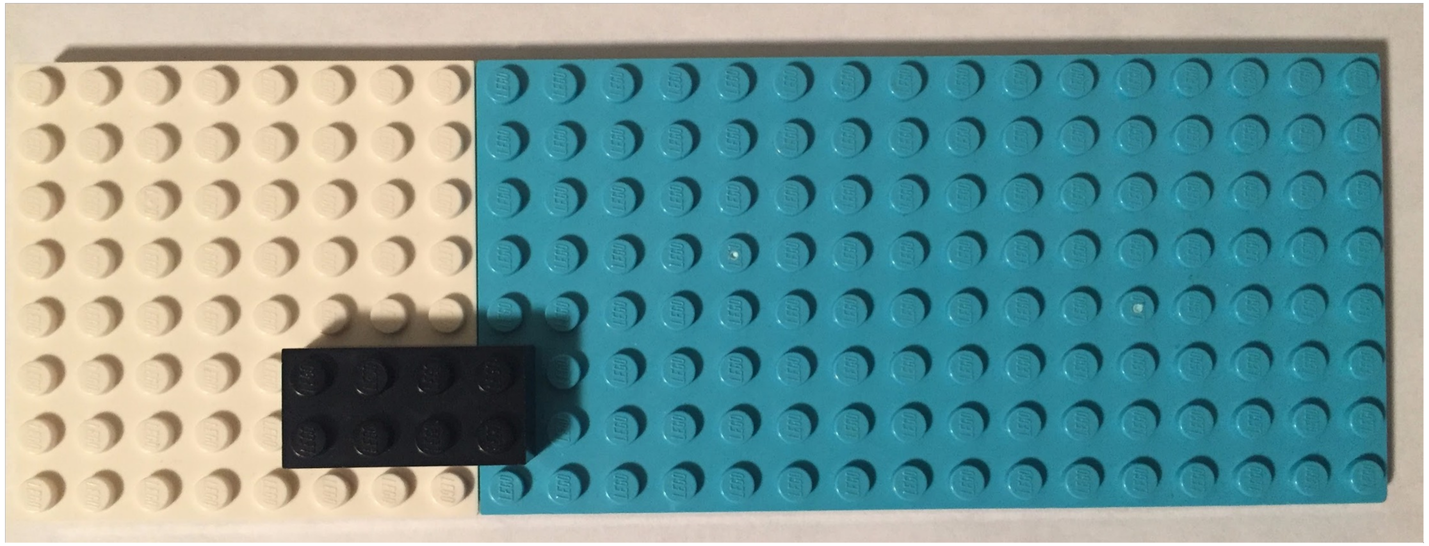
Bayes' theorem

$$P(\text{white}|\text{black}) = \frac{P(\text{black}|\text{white}) * P(\text{white})}{P(\text{black})}$$



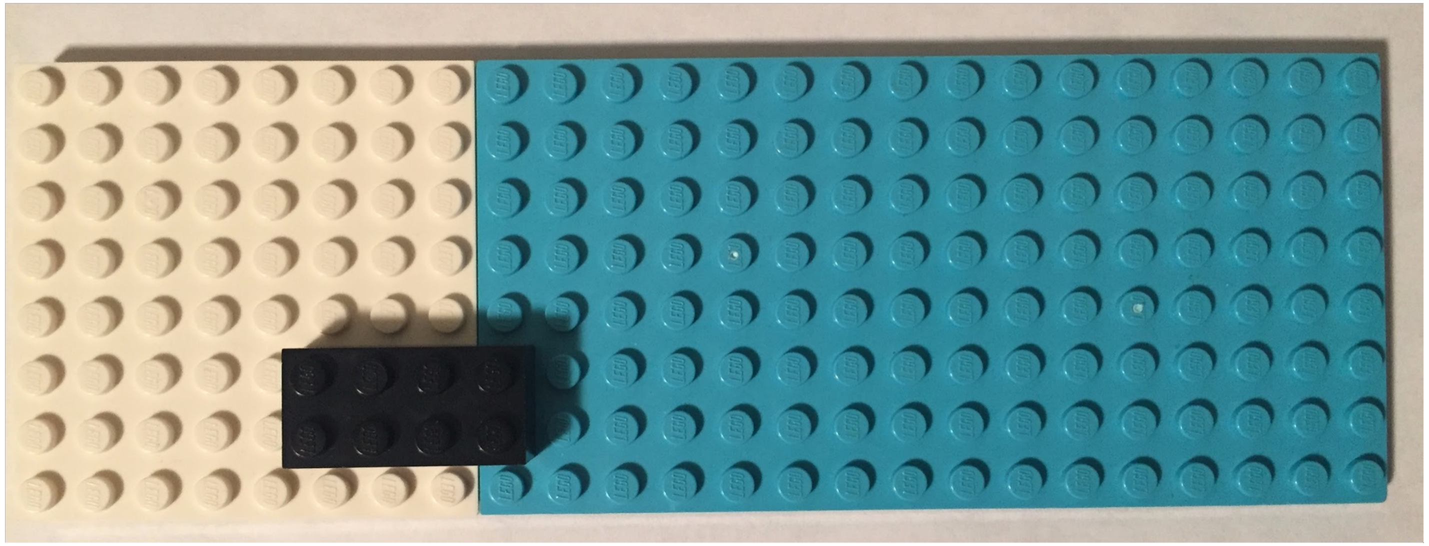
Bayes' theorem

$$P(\text{white}|\text{black}) = \frac{0.09375 * 0.33}{0.04165}$$

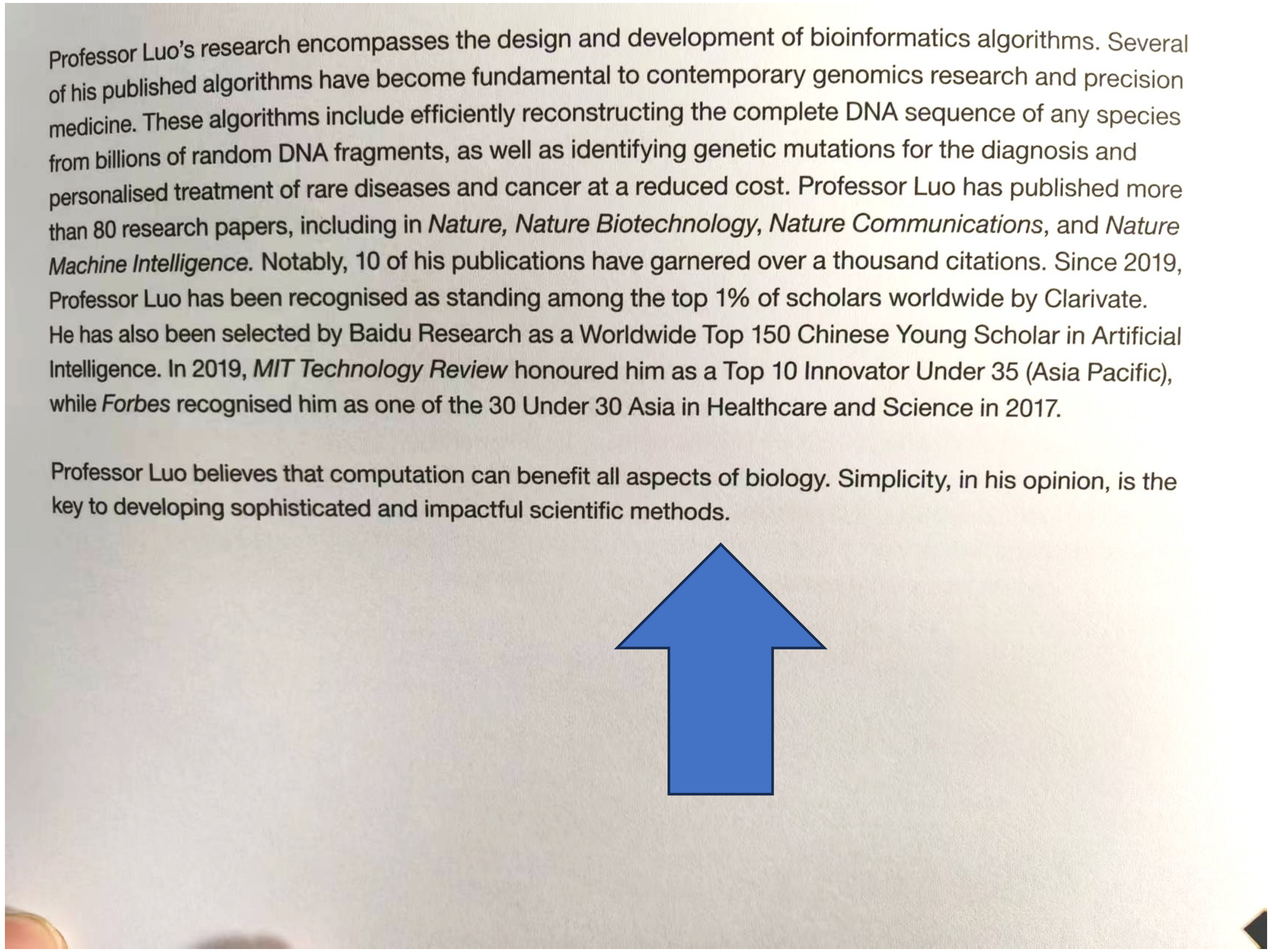


Bayes' theorem

P(white|black) = 0.75



While we can intuit these probabilities spatially with Legos, the beauty of Bayes' theorem is that it can be generalized to situations that we cannot easily intuit



Professor Luo's research encompasses the design and development of bioinformatics algorithms. Several of his published algorithms have become fundamental to contemporary genomics research and precision medicine. These algorithms include efficiently reconstructing the complete DNA sequence of any species from billions of random DNA fragments, as well as identifying genetic mutations for the diagnosis and personalised treatment of rare diseases and cancer at a reduced cost. Professor Luo has published more than 80 research papers, including in *Nature, Nature Biotechnology*, *Nature Communications*, and *Nature Machine Intelligence*. Notably, 10 of his publications have garnered over a thousand citations. Since 2019, Professor Luo has been recognised as standing among the top 1% of scholars worldwide by Clarivate. He has also been selected by Baidu Research as a Worldwide Top 150 Chinese Young Scholar in Artificial Intelligence. In 2019, *MIT Technology Review* honoured him as a Top 10 Innovator Under 35 (Asia Pacific), while *Forbes* recognised him as one of the 30 Under 30 Asia in Healthcare and Science in 2017.

Professor Luo believes that computation can benefit all aspects of biology. Simplicity, in his opinion, is the key to developing sophisticated and impactful scientific methods.

Applications of Bayes theorem

- Widely used in machine learning and finance
- Decision making in driverless cars
- Email spam detection
- Assess disease risk from test results
- Voice recognition
- Text autocompletion

…

More info:

https://arbital.com/p/bayes_rule/?l=1zq
https://betterexplained.com/articles/an-intuitive-and-short-explanation-of-bayes-theorem/

Bayes theorem example

Suppose a test for COVID-19 has an accuracy of 99%, + denotes a positive test and *D* represents a disease (1) or not (0)

$$\mathrm{Prob}(+ \mid D=1) = 0.99, \mathrm{Prob}(- \mid D=0) = 0.99$$

Suppose we select a random person and they test positive. What is the probability that they have the disease? We write this as $\mathrm{Prob}(D=1 \mid +)$
Let's assume the COVID-19 rate is 1 in 4,000 which implies that $\mathrm{Prob}(D=1) = 0.00025$

We solve the problem as

$$\begin{aligned}\mathrm{Pr}(D=1 \mid +) &= \frac{P(+ \mid D=1) \cdot P(D=1)}{\mathrm{Pr}(+)} \\ &= \frac{\mathrm{Pr}(+ \mid D=1) \cdot P(D=1)}{\mathrm{Pr}(+ \mid D=1) \cdot P(D=1) + \mathrm{Pr}(+ \mid D=0)\mathrm{Pr}(D=0)}\end{aligned}$$

Bayes theorem example

Plugging in the numbers

$$\frac{0.99 \cdot 0.00025}{0.99 \cdot 0.00025 + 0.01 \cdot (.99975)} = 0.024158$$

This says that despite the test having 0.99 accuracy, the probability of having the disease given a positive test is only 0.024. This may appear counter-intuitive to some, but the reason this is the case is because we have to factor in the very rare probability that a person, chosen at random, has the disease

Visualize the example with Monte Carlo Simulation

```
R 4.2.2 · ~/
> prevalance <- 0.00025
> N <- 100000
> outcome <- sample(c("Disease","Healthy"), N, replace = TRUE,
+                   prob = c(prevalance, 1 - prevalance))
```

We start by randomly selecting 100,000 people from a population in which the disease in question has a 1 in 4,000 prevalence

```
R 4.2.2 · ~/
> N_D <- sum(outcome == "Disease")
> N_D
[1] 23
> N_H <- sum(outcome == "Healthy")
> N_H
[1] 99977
> #> [1] 99977
```

Note that there are very few people with the disease

Visualize the example with Monte Carlo Simulation

R 4.2.2 · ~/

```
> accuracy <- 0.99
> test <- vector("character", N)
> test[outcome == "Disease"]  <- sample(c("+", "-"), N_D, replace = TRUE,
+                                       prob = c(accuracy, 1 - accuracy))
> test[outcome == "Healthy"]  <- sample(c("-", "+"), N_H, replace = TRUE,
+                                       prob = c(accuracy, 1 - accuracy))
```

Also, there are many without the disease, which makes it more probable that we will see some false positives given that **the test is not perfect.** Now each person gets the test, which is correct 99% of the time

R 4.2.2 · ~/

```
> table(outcome, test)
         test
outcome       -     +
  Disease     1    22
  Healthy 98933  1044
```

Because there are so many more controls than cases, even with a low false positive rate we get more controls than cases in the group that tested positive

From this table, we see that the proportion of positive tests that have the disease is 22 out of 1044. We can run this over and over again to see that, in fact, the probability converges to about 0.024

It is observed that students who liked a course responded fewer than those who disliked. It is not uncommon but is understandable for underperforming students to use SFTL as a “retaliation”. However, the better a course, the more it will be underrated because of this rater bias. Let’s see why as a data scientist.

A: students who liked the course

B: students who rated

$$P(A \mid B) = \frac{P(B \mid A) \times P(A)}{P(B)}$$

$$P(A \mid B) = \frac{0.25 \times 0.9}{0.5} = 0.45$$

Let’s say P(A) is 0.9 😝

Let’s say P(B) is 0.5 (historical average of all students)

P(A | B) is the SFTL result

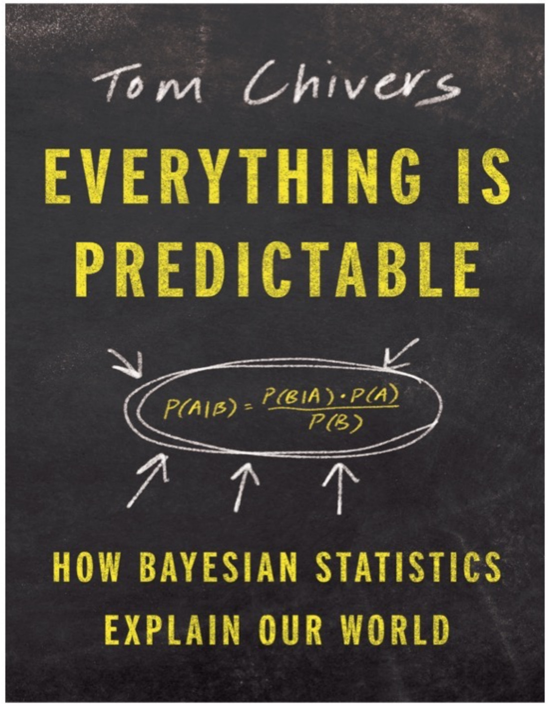
P(B | A) is the rate of those who liked my course rated the course

If P(B | A) is only 25%, which is lower than the average response rate P(B), P(A | B) is only 0.45, that’s a 0.45 drop from the reality P(A), and the higher the P(A), the higher the drop 😱

So, if you liked the course, and if you want to see more courses provided by HKU to share real experiences and follow the latest technological developments, please give me a good rating in SFTL, THANK YOU!

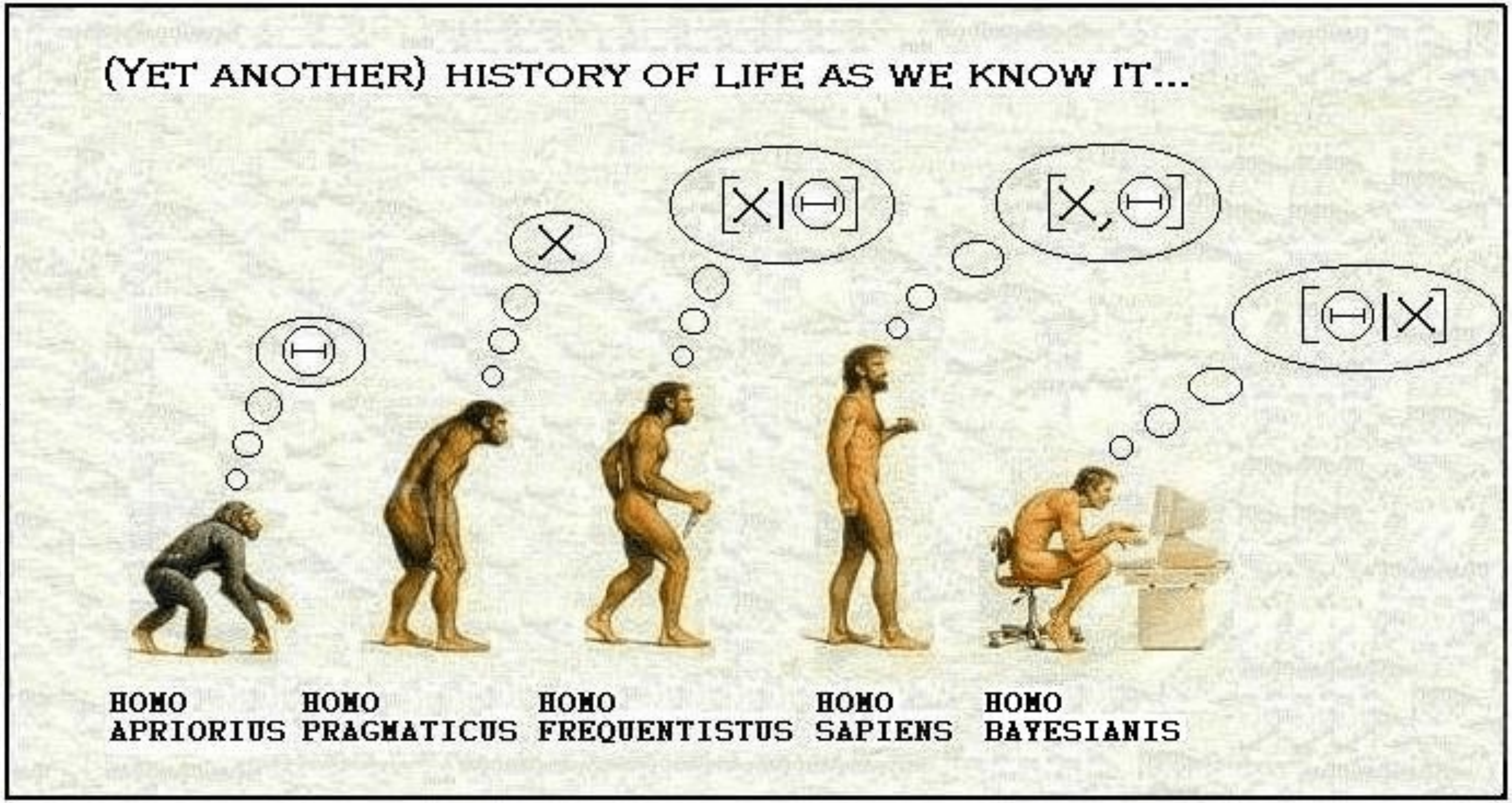
- If you are interested, read textbook chapter 16.8 Election forecasting using Bayes theorem, and the backgrounds in 16.1 and 16.2

  http://rafalab.dfci.harvard.edu/dsbook/



Everything is predictable by Tom Chivers

This is a great book introducing Bayesian statistics. If you are interested, buy a copy or ask the lecturer for a PDF preview for educational purposes.



Reality does not matter, observation is the king	Observation does not matter, previous reality means truth	Measures the probability of reality given observation	Your observation is not necessarily my reality	Measures the probability of observation given previous reality

https://pactiss.org/2011/11/02/bayesian-inference-homo-bayesianis/